SUMMARY TABLE: ESSENTIAL MATHEMATICS GRADE 10

Sub-Strand 1.3: Quadratic Equations — Teacher Reference (Pre-filled)

SUMMARY TABLE: ESSENTIAL MATHEMATICS GRADE 10	
Sub-Strand	Sub-Strand 1.3: Quadratic Equations
Driving Question	DRIVING QUESTION: How can writing and solving a quadratic equation help us find unknown dimensions, quantities, and arrangements in real-life situations across Kenya?

INSTRUCTIONS
FOR TEACHERS: This is the teacher reference version — each row is pre-filled with expected student responses. Use it to assess student Summary Tables, identify gaps, and prepare discussion questions. The student version has blank cells for students to complete after each lesson.

Lesson # and Title	What did I observe?	What did I learn?	How does this explain the phenomenon?
Lesson 1 **Anchoring Phenomenon & Forming Algebraic Expressions**	Teacher presents the anchoring phenomenon: Achieng is arranging chairs in a hall in Kisumu for a harambee event. The number of columns exceeds the number of rows by 7, and the total number of chairs is 120. Students observe that two unknown but related quantities (rows and columns) multiply to give a known product. Teacher should display a physical or drawn grid of chairs and ask students to predict the number of rows before any instruction. Cold-call at least 3 students to share predictions and record these on the Driving Question Board (DQB). Allow 30 seconds of wait time after the prompt: 'What do you notice about the relationship between the rows and columns?'	Students learn to represent unknown quantities algebraically: if the number of rows is r, then the number of columns is $r + 7$. This is a foundational skill — teacher should emphasise that choosing the smaller or more basic unknown as the variable (r) simplifies the expression for the related quantity. The product of the two expressions gives the total: $r(r + 7) = 120$. Pedagogical note: Ensure students understand that r represents a specific, real number of rows — it is not arbitrary. Reinforce that the letter is a placeholder for a value we are trying to find.	Students explain that when one unknown exceeds another by a fixed amount and their product is known, we can let the smaller quantity equal r and write the larger as $r + 7$. Multiplying these gives $r(r + 7) = 120$. Teacher should confirm that this single equation holds all the information needed to solve the problem — both unknowns are expressed in terms of one variable. Connect back to the phenomenon: 'We now have a mathematical sentence that describes Achieng's chair arrangement. Our next step is to learn how to solve it.'
Lesson 2 **From Expressions to Quadratic Expressions: Recognising and Writing $ax^2 + bx + c$**	Teacher guides students to expand the expression $r(r + 7)$ by distributing: $r \times r = r^2$, and $r \times 7 = 7r$, giving $r^2 + 7r$. Students observe that multiplying two linear expressions — one with degree 1 and one with degree 1 — always produces a term with degree 2 (the squared term). Teacher provides 3–4 additional pairs of linear	Students learn that the product of two linear expressions always yields a quadratic expression of the standard form $ax^2 + bx + c$, where $a \neq 0$. The x^2 (or r^2) term arises because $x \times x = x^2$. Teacher should use colour-coding on the board: highlight the squared term in one colour, the linear term in another, and the constant in a third.	Students explain that expanding $r(r + 7)$ gives $r^2 + 7r$, and that substituting the known product (120) produces $r^2 + 7r = 120$, which rearranges to $r^2 + 7r - 120 = 0$. This is now in standard quadratic form $ax^2 + bx + c = 0$, where $a = 1$, $b = 7$, and $c = -120$. Teacher should connect explicitly to the anchoring phenomenon: 'Every time we

	expressions for students to expand (e.g., from Kenyan contexts: length and width of a maize farm plot near Eldoret, sides of a rectangular fish pond at Lake Victoria). Ask students: 'What do all the expanded results have in common?' Allow 20 seconds of wait time, then cold-call 3 students.	Pedagogical note: Students frequently forget to multiply all terms when expanding — teacher should model the 'each term in the first bracket multiplies each term in the second bracket' rule explicitly, using arrows or the FOIL mnemonic as a scaffold.	expand Achieng's row-and-column expressions, we get a quadratic equation. This is why the equation type is called quadratic — from the Latin for square, because area (and arrangements) involve squared terms.'
Lesson 3 **Forming Quadratic Equations from Real-Life Situations**	Teacher presents 2–3 new Kenyan real-life scenarios that generate quadratic equations: e.g., a tea farmer in Kericho whose rectangular plot has a length 5 m more than its width and an area of 84 m²; a matatu operator in Nairobi whose daily passengers exceed daily trips by 4 and whose product is 96. Students observe each scenario, identify the two related unknowns, and attempt to write the equation before instruction. Teacher circulates and notes common errors (e.g., students writing $x + 5 = 84$ instead of $x(x + 5) = 84$). Allow 1 minute of independent thinking before pair discussion. Cold-call 4 pairs to share their equations.	Students learn the three-step process for forming a quadratic equation from a real-life situation: (1) identify the two related unknowns and define a variable for the smaller/simpler one; (2) express the second unknown in terms of the first; (3) write the product (or other relationship) equal to the known total and rearrange to standard form $ax^2 + bx + c = 0$. Pedagogical note: Stress that the equation must equal zero in standard form before any solving method is applied. Students often stop at $x(x + 5) = 84$ — teacher should demonstrate rearranging to $x^2 + 5x − 84 = 0$ and explain why this form is necessary for the factorisation methods ahead.	Students explain that whenever two related unknown quantities are multiplied to give a known total, the resulting equation always contains a squared term and is therefore quadratic. They can now form quadratic equations from diverse Kenyan contexts — farming plots, event seating, market quantities — and write them in standard form. Teacher should ask: 'How does every equation we formed today link back to Achieng's chair problem in Kisumu?' Expected student response: both involve two related unknowns whose product is known, producing the same algebraic structure.
Lesson 4 **Factorisation of Quadratic Expressions: The Product-and-Sum Method**	Teacher writes $x^2 + 7x − 120$ on the board (Achieng's expression) and asks: 'Can we reverse the expansion? Can we write this as a product of two brackets?' Students attempt to guess the brackets before instruction. Teacher then introduces the product-and-sum method with a structured example: find two numbers whose product = −120 (the constant term, c) and whose sum = +7 (the coefficient of x, b). Teacher provides a factor-pair table for students to complete collaboratively for the numbers −120 and +7. Allow 2 minutes of pair work. Cold-call 3 pairs to report candidate factor pairs.	Students learn the product-and-sum method: for a quadratic $x^2 + bx + c$, find integers p and q such that $p × q = c$ and $p + q = b$, then factorise as $(x + p)(x + q)$. For $x^2 + 7x − 120$, the correct pair is $p = 15$, $q = −8$ (since $15 × −8 = −120$ and $15 + (−8) = 7$), giving $(x + 15)(x − 8)$. Pedagogical note: Many students struggle with negative constants. Teacher should dedicate time to the sign rules: if c is negative, the two numbers have opposite signs; if c is positive and b is positive, both numbers are positive; if c is positive and b is negative, both numbers are negative. Use a simple sign-decision chart as a classroom poster.	Students explain that the product-and-sum method reverses expansion by identifying the two numbers that were originally added (to give bx) and multiplied (to give c) inside the brackets. For Achieng's equation, $x^2 + 7x − 120 = (x + 15)(x − 8)$. Teacher should verify by re-expanding: $(x + 15)(x − 8) = x^2 − 8x + 15x − 120 = x^2 + 7x − 120$ ✓. Reinforce that factorisation is a tool — its power is revealed in the next lesson when we use it to solve the equation and find the actual number of rows in Achieng's hall.
Lesson 5 **Solving Quadratic Equations by Factorisation: Finding and Interpreting Both Roots**	Teacher writes the fully factorised equation $(x + 15)(x − 8) = 0$ on the board and asks: 'If two numbers multiply to give zero, what must be true about at least one of them?' Allow 20 seconds of wait time. Cold-call 3 students. Teacher then demonstrates: if $(x + 15)(x − 8) = 0$, then either $x + 15 = 0$,	Students learn the Zero Product Property: if $A × B = 0$, then $A = 0$ or $B = 0$ (or both). This allows each factor to be set equal to zero independently, yielding two roots. They also learn that real-life context governs which root(s) are valid: a negative number of rows is impossible, so $x = −15$ is rejected	Students explain that solving by factorisation uses the Zero Product Property to find two roots, and that context determines which root(s) are meaningful. For Achieng's problem, the hall has 8 rows and 15 columns of chairs. Teacher should connect this full solution back to the driving

	giving $x = -15$, or $x - 8 = 0$, giving $x = 8$. Students observe that a quadratic equation always has two roots. Teacher asks: 'Achieng is arranging real chairs in a real hall. Which root makes sense — $x = -15$ or $x = 8$? Why?' Allow 30 seconds of think time before cold-calling.	and $x = 8$ (rows) is accepted. The number of columns is then $8 + 7 = 15$. Verification: $8 \times 15 = 120$ ✓. Pedagogical note: Insist that students always check both roots against the context and write a conclusion sentence. This is a key mathematical practice — sense-making beyond the algebra.	question: 'We started with a real arrangement problem in Kisumu, wrote an algebraic expression, formed a quadratic equation, factorised it, and used the roots to find the actual dimensions. That is the full power of quadratic equations in real life.'
Lesson 6 **Solving Quadratic Equations by Completing the Square**	Teacher presents a quadratic equation that does not factorise neatly with integers (e.g., $x^2 + 6x - 7 = 0$, or a Kenyan context: a rectangular shamba in Nakuru with area 55 m^2 where the length exceeds the width by 6 m, giving $x^2 + 6x - 55 = 0$). Teacher first asks students to attempt factorisation and observes that some equations require an alternative method. Teacher then introduces completing the square: rewrite $x^2 + bx$ as $(x + b/2)^2 - (b/2)^2$. Demonstrate step by step on the board, inviting students to predict each next step. Cold-call 3 students at each stage. Allow 30 seconds of wait time after posing: 'What do we add to both sides to create a perfect square?'	Students learn the completing-the-square method: (1) ensure the equation is in the form $x^2 + bx + c = 0$ (coefficient of x^2 is 1); (2) move the constant to the right side; (3) add $(b/2)^2$ to both sides to form a perfect square trinomial; (4) write the left side as $(x + b/2)^2$; (5) take the square root of both sides (remembering ±); (6) solve for x. Pedagogical note: Students commonly forget the ± when taking the square root — teacher should highlight this explicitly and explain that it is the source of the two roots. Also ensure students simplify surds or decimals as appropriate, and always check solutions in the original equation.	Students explain that completing the square is a universal method for solving any quadratic equation of the form $x^2 + bx + c = 0$, even when factorisation with integers is not possible. The method transforms the quadratic into a perfect square expression, making it solvable by taking a square root. For Kenyan real-life problems involving area, distance, or quantity, completing the square allows learners to find unknown dimensions precisely. Teacher should connect to the anchoring phenomenon: 'If Achieng's product had not been 120 — say the total chairs were 130 — the equation might not factorise neatly, and completing the square would be our reliable tool.'
Lesson 7 **Applications and Digital Research: Quadratic Equations in the Real World**	Teacher facilitates a structured research and application session. Students (in groups) explore real Kenyan contexts where quadratic equations arise: seating arrangements for county functions, rectangular agricultural plots in the Rift Valley, profit-maximisation for a mandazi seller in Mombasa, projectile paths in athletics (inspired by Kenyan runners' training drills). Each group is given a scenario card with a quadratic equation to form and solve. Groups also use available digital tools (if accessible) to verify solutions graphically. Teacher circulates, prompts with: 'Where is the quadratic structure in your scenario — what two related unknowns are being multiplied?' Allow 10 minutes of group work before presentations.	Students learn that the quadratic structure first seen in Achieng's chair problem — (unknown quantity) × (unknown quantity + fixed difference) = known total — recurs across many real-life Kenyan situations: area of rectangular land, total items in a rectangular arrangement, revenue calculations (price × quantity where both relate to the same variable), and even motion problems. Pedagogical note: Emphasise transferability — the algebraic method (form → expand → rearrange → factorise or complete the square → interpret roots in context) is identical regardless of the scenario. Students should see the method as a general problem-solving tool, not a one-off procedure tied only to chairs.	Students explain how to identify a real-life situation that generates a quadratic equation, form the equation correctly, solve it using an appropriate method (factorisation or completing the square), and interpret the roots in context. Teacher should ask each group: 'How does your scenario connect to our driving question?' Expected response: finding unknown dimensions, quantities, or arrangements in real Kenyan situations is exactly what quadratic equations help us do. Record new insights on the DQB and celebrate examples that extend beyond the original anchoring phenomenon.
Lesson 8 **Model Building, Consolidation & Final**	Teacher returns to the original anchoring phenomenon — Achieng's 120-chair arrangement in Kisumu — and displays the	Students consolidate the full solution pathway for a quadratic equation arising from a real-life situation: (1) define the	Students explain that a quadratic equation of the form $x^2 + 7x - 120 = 0$ arises when two related unknown dimensions (such as

Explanation: The Full Journey of a Quadratic Equation	full DQB, showing how student questions and understanding have evolved across all 8 lessons. Students review their initial predictions from Lesson 1 and compare them with their current knowledge. Teacher asks: 'At the start, how did you try to find the number of rows? How do you solve this now?' Cold-call 4 students. Students then individually attempt a parallel problem (e.g., a school in Eldoret arranges 180 desks in rows where columns exceed rows by 3 — find the number of rows), applying the complete method from start to finish. Allow 8 minutes of independent work.	variable for the simpler unknown; (2) express the related unknown in terms of the variable; (3) write the product/relationship equal to the known total; (4) expand and rearrange to standard form $ax^2 + bx + c = 0$; (5) factorise using the product-and-sum method (or complete the square if needed); (6) apply the Zero Product Property to find both roots; (7) interpret roots in context and reject any that are not valid (e.g., negative dimensions); (8) verify the solution. Pedagogical note: This eight-step pathway should be displayed as a permanent classroom anchor chart. Encourage students to articulate each step verbally — mathematical communication is a core CBE competency.	rows and columns of a chair arrangement) have a known product and a known difference. It is solved by factorising into $(x + 15)(x - 8) = 0$, applying the Zero Product Property to get $x = -15$ or $x = 8$, and then rejecting $x = -15$ because a negative number of rows has no physical meaning in Achieng's hall. The valid solution, $x = 8$ rows and 15 columns, satisfies $8 \times 15 = 120$ ✓. Teacher closes by returning to the driving question: 'How can writing and solving a quadratic equation help us find unknown dimensions, quantities, and arrangements in real-life situations across Kenya?' Students should now be able to answer this question fully, fluently, and with examples drawn from their own Kenyan contexts.